

Bachelor of Science (Honours) in Data Science and Artificial Intelligence

DA 102 – First Lab



Basic Spread Sheet Modeling



Problem 1



- In DA902 course, the instructor gave **FIVE** graded homework assignments and three graded exams
- Each homework carries maximum of 25 marks
- Each exam carries maximum of 100 marks.
- All five homeworks' carry 25% weightage
- All three exams' carry 75% weightage
- Marks obtained by each student in homework and exam are given in the excel sheet.
- Your task is to compute Homeworks weightage and exams weightage and total marks (sum of homework weightage and exam weightage)

Problem 2



- You are given the following
 - Person ID
 - Person's height in centimeters
 - Person's weight in grams
- Your task is to compute Body Mass Index (BMI) given by the formula: $703 * w / (h * h)$.
Where w is weight of person in kilograms and h is height of person in meters.

Problem 3



- The famous butterfly effect states that if a butterfly flaps its wings in Tahiti (an island in the S Pacific, in the Windward group of the Society Islands)
- That small event might cause a hurricane to hit Texas
- Suppose the weather at time t is always between 0 and 1 and is governed by the equation
- Person ID $x_{t+1} = 4 * x_t * (1 - x_t)$
- Given $x_{t=0} = 0.3$ compute $x_1, x_2, x_3, \dots, x_{50}$
- Given $x_{t=0} = 0.300001$ compute $x_1, x_2, x_3, \dots, x_{50}$
- How your calculations illustrate butterfly effect?

Problem 4



- A lake is currently stocked with 12230 fish
- Each year, the number of births per fish is 1.2
- Each year, the number of deaths per fish is 0.7
- Show that if 6115 fish per year are harvested, the number of fish in the lake will stay constant
- To achieve this task, develop formulas for columns in the excel sheet week01-problem4
 - Initial number of fish
 - Births
 - Deaths
 - Harvested fish
 - Fish at the end of the year



Thank You!

